19/08/2026

CODE-B
Phase-2

Aakash
Medical | IIT-JEE | Foundations

Corporate Office: AESL, 3rd Floor, Incuspaze Campus-2, Plot no. 13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

FORTNIGHTLY TEST FOR NEET-2027 (XII PASSED) PH-2

MM: 720 **FT-03B** **Time: 180 Mins.**

Answers

1. (3)	37. (4)	73. (4)	109. (3)	145. (3)
2. (3)	38. (1)	74. (3)	110. (1)	146. (2)
3. (3)	39. (1)	75. (4)	111. (3)	147. (2)
4. (1)	40. (1)	76. (2)	112. (3)	148. (4)
5. (2)	41. (2)	77. (1)	113. (4)	149. (3)
6. (2)	42. (2)	78. (4)	114. (4)	150. (4)
7. (3)	43. (3)	79. (1)	115. (3)	151. (1)
8. (4)	44. (1)	80. (2)	116. (2)	152. (2)
9. (3)	45. (1)	81. (2)	117. (3)	153. (2)
10. (3)	46. (4)	82. (2)	118. (1)	154. (1)
11. (2)	47. (3)	83. (4)	119. (2)	155. (3)
12. (4)	48. (3)	84. (1)	120. (2)	156. (1)
13. (3)	49. (2)	85. (4)	121. (3)	157. (3)
14. (3)	50. (1)	86. (2)	122. (2)	158. (2)
15. (2)	51. (1)	87. (2)	123. (2)	159. (3)
16. (3)	52. (4)	88. (3)	124. (4)	160. (1)
17. (2)	53. (2)	89. (4)	125. (4)	161. (1)
18. (3)	54. (4)	90. (2)	126. (2)	162. (4)
19. (3)	55. (4)	91. (4)	127. (3)	163. (2)
20. (4)	56. (1)	92. (4)	128. (4)	164. (3)
21. (3)	57. (1)	93. (2)	129. (1)	165. (1)
22. (4)	58. (3)	94. (3)	130. (3)	166. (2)
23. (2)	59. (3)	95. (4)	131. (3)	167. (4)
24. (3)	60. (4)	96. (1)	132. (3)	168. (3)
25. (2)	61. (1)	97. (3)	133. (1)	169. (3)
26. (4)	62. (4)	98. (2)	134. (2)	170. (3)
27. (2)	63. (3)	99. (4)	135. (3)	171. (3)
28. (2)	64. (2)	100. (3)	136. (2)	172. (3)
29. (4)	65. (3)	101. (3)	137. (1)	173. (4)
30. (3)	66. (2)	102. (3)	138. (3)	174. (2)
31. (2)	67. (4)	103. (4)	139. (4)	175. (2)
32. (4)	68. (3)	104. (3)	140. (2)	176. (3)
33. (1)	69. (1)	105. (4)	141. (3)	177. (1)
34. (2)	70. (1)	106. (4)	142. (1)	178. (2)
35. (1)	71. (3)	107. (2)	143. (4)	179. (4)
36. (2)	72. (3)	108. (3)	144. (3)	180. (1)

Hints and Solutions

PHYSICS

(1) **Answer :** (3)
Solution:
Only component of velocity normal to surface, changes.
$e = \frac{v_y}{3} = \frac{1}{3}$
$\Rightarrow v_y = 1$
$\therefore \overrightarrow{v}_f = 2\hat{i} + \hat{j}$

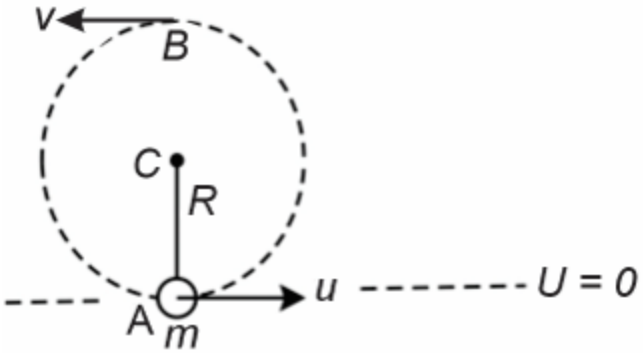
(2) **Answer :** (3)
Solution:
P.E. (U) = $\frac{1}{2}kx^2$
$U_i = \frac{1}{2}k.(2)^2$
$U_f = \frac{1}{2}K(8)^2$
$U_f = 16\,U$
Increase $\Delta U = U_f - U_i = 15\,U$

(3) **Answer :** (3)
Solution:
Centripetal force is perpendicular to instantaneous velocity at every instant of motion therefore $dW = \overrightarrow{F} \cdot \overrightarrow{ds} = F \cdot d\cos\theta = 0$
$(\because \ \theta = 90^\circ,\ \cos 90^\circ = 0)$

(4) **Answer :** (1)
Solution:
$P = \frac{mgh}{t} = \frac{800\times 10\times 10}{80} = 1000$ J
If $\chi = 75\%$
$\frac{3}{4}x = 1000$
$x = \frac{4000}{3} \simeq 1300$ W

(5) **Answer :** (2)
Solution:
$P = \frac{dk}{dt}$
$P = \frac{d}{dt}\left(9t^2\right)$
$P = 18t$
Power will increases with time

(6) **Answer :** (2)
Solution:



By conservation of energy

$K_A + U_A = K_B + U_B$
$\frac{m}{2}u^2 + 0 = \frac{m}{2}v^2 + mg(2R)$
$K_A = \frac{mu^2}{2} = \frac{m}{2}\left(3\sqrt{gR}\right)^2 + mg\left(2R\right)$
$K_B = \frac{m}{2}\left(3\sqrt{gR}\right)^2$
$\frac{K_A}{K_B} = \frac{13}{9}$

(7) **Answer :** (3)
Solution:
Using conservation of energy,
$\frac{1}{2}m(0)^2 + mg\left(7\right) = \frac{1}{2}mv^2$
$\Rightarrow \ (K.E.)_R = \frac{1}{2}mv^2 = mg\left(7\right) = 2 \times 10 \times 7 = 140 \text{ J}$

(8) **Answer :** (4)
Solution:
$v' = 0.6\sqrt{2gh}$
$h' = \frac{v^2}{2g} = \frac{(0.6)(0.6)2gh}{2g}$
$h' = 0.36 \times 1 \text{ m}$
= 0.36 m

(9) **Answer :** (3)
Solution:
In collision net linear momentum of the system remains conserved.
$\left|\overrightarrow{P}_i\right| = \left|\overrightarrow{P}_f\right|$
$\frac{1}{4} \times 3 - \frac{1}{4} \times 1 = \left(\frac{1}{4} + \frac{1}{4}\right) \times v$
$\frac{1}{2} = \frac{1}{2} \times v = v = 1 \text{ m s}^{-1}$

(10) **Answer :** (3)
Solution:
Let $P_1 = P = mv$, $v_1 = v$
$P_2 = 1.2\,mv$, $v_2 = 1.2v$
$\frac{\Delta K}{K} \times 100 = \frac{\frac{1}{2}m\left[(1.2v^2) - v^2\right]}{\frac{1}{2}mv^2} \times 100\%$
$= \frac{\frac{1}{2}mv^2[1.44 - 1]}{\frac{1}{2}mv^2} \times 100\% = 44\%$

(11) **Answer :** (2)
Solution:
According to conservation of momentum
2(6) + 4(0)= 2(0)+4(v)
12 = 4 v
v = 3 m/s
$\text{KE}_\text{i} = \frac{1}{2}(2)\,(6)^2 + \frac{1}{2}(4)\,(0) = 36 \text{ J}$
$\text{KE}_\text{f} = \frac{1}{2}(2)\,(0)^2 + \frac{1}{2}(4)\,(3)^2 = 18 \text{ J}$
Loss of KE of system = 36 – 18 = 18 J

(12) **Answer :** (4)
Solution:
Work done by friction may be zero, positive or negative. Body at rest can have mechanical energy due to its potential energy.

(13) **Answer :** (3)
Solution:
$W = \int F \cdot dx = \int_0^3 (10 + 2x)\,dx = \left[10x + x^2\right]_0^3$
= 39 Joule

(14) **Answer :** (3)
Solution:

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$\frac{1}{2}kx^2 = \frac{1}{2}mv^2$

$x = v\sqrt{\frac{m}{K}}$

(15) Answer : (2)
Solution:
Work done by conservative force = $-\Delta U$ = positive
Hence potential energy should decrease.

(16) Answer : (3)
Solution:

$F = k_1x_1 = k_2x_2 \qquad k_1 > k_2 \Rightarrow \frac{k_2}{k_1} < 1$

$\frac{W_1}{W_2} = \frac{\frac{1}{2}k_1x_1^2}{\frac{1}{2}k_2x_2^2} = \frac{(k_1x_1)x_1}{(k_2x_2)x_2} = \frac{x_1}{x_2} = \frac{k_2}{k_1} < 1$

$\Rightarrow \frac{W_1}{W_2} < 1 \Rightarrow W_1 < W_2$

(17) Answer : (2)
Solution:
Work done will be equal to
Area under force-displacement curve
$10 \times 10 + \frac{1}{2} \times 10 \times 10 = 150$ J

(18) Answer : (3)
Solution:
In stable equilibrium, the potential energy is minimum.

(19) Answer : (3)
Solution:

$F = -\frac{dU}{dx} = -\left[\frac{d}{dx}\left(ax^2 - bx\right)\right] = -[2ax - b]$

= –2*ax* + *b*

(20) Answer : (4)
Solution:
By using work energy theorem
$\Delta K = \Delta W$
$\therefore \quad \Delta K = F \times \Delta x = \frac{1}{2}(5^2 - 5^2) =$ Zero

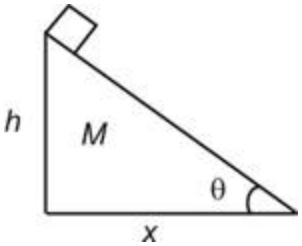
(21) Answer : (3)
Solution:
Linear momentum of the system will remain conserved

$p^2 = p_1^2 + p_2^2 + 2p_1p_2\cos\theta$

$\left\{2m\frac{v}{2}\right\}^2 = \left(mv\right)^2 + \left(mv\right)^2 + 2\left(mv\right)\left(mv\right)\cos\theta$

$\Rightarrow \theta = 120°$

(22) Answer : (4)
Solution:



$\frac{b}{x} = \tan\theta$
$x = h\cot\theta$
$\because F_x = 0 \quad \Delta x_{cm} = 0$
$0 = m(h\cot\theta - x) - Mx$
$x = \frac{mh\cot\theta}{m+M}$

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(23) **Answer :** (2)
Solution:
$I_1\omega_1 = I_2\omega_2$
$Mr^2 \times \omega = r^2\,[2m + M]\ \times\ \omega'$
$\omega'\ = \frac{\omega M}{(M+2m)}$

(24) **Answer :** (3)
Hint:
Use appropriate formulae.
Solution:

	(Quantity)		**(Corresponding formula)**
A.	Torque τ	II.	$I\,\alpha$
B.	Angular momentum	III.	$Mvr_{\perp}$
C.	$\sum \vec{F}_{\text{ext}}$	I.	$M\,\vec{a}_{\text{cm}}$
D.	Instantaneous power	IV.	$\vec{\tau}\,.\,\vec{\omega}$

(25) **Answer :** (2)
Solution:
$\frac{I_1}{I_2} = \frac{m_1 r_1^2}{m_2 r_2^2} = \frac{2}{1}$
$\Rightarrow\ \frac{m_1}{m_2} \times \left(\frac{2}{1}\right)^2 = \frac{2}{1}$
$\Rightarrow\ \frac{m_1}{m_2} = \frac{1}{2}$

(26) **Answer :** (4)
Solution:
$\frac{2}{5}MR^2 + MR^2 = MK^2$
$K = R\sqrt{\frac{7}{5}}$

(27) **Answer :** (2)
Solution:
$\frac{K'-K}{K} \times 100 = 800\,\%\quad \Rightarrow\quad K' = 9\,K$
$K = \frac{L^2}{2I}\ \Rightarrow\ L\ \propto\ \sqrt{K}$
$\frac{L'-L}{L} \times 100 = \frac{3\sqrt{K}-\sqrt{K}}{\sqrt{K}} \times 100 = 200\%$

(28) **Answer :** (2)
Solution:
If $\vec{F}_{ext} = 0 \Rightarrow \vec{a}_{cm} = 0$
Also, $\vec{F}_{ext} = \left(\frac{\Delta \vec{P}}{\Delta t}\right)_{\text{system}}$
$\therefore$ If $\vec{F}_{\text{ext}} \Rightarrow \left(\Delta \vec{P}\right) = 0$

(29) **Answer :** (4)
Solution:
$I = \frac{ML^2}{3} + ML^2$
= $\frac{4ML^2}{3}$

(30) **Answer :** (3)
Solution:
If a particle moves in a straight line with constant velocity then its angular momentum is constant.

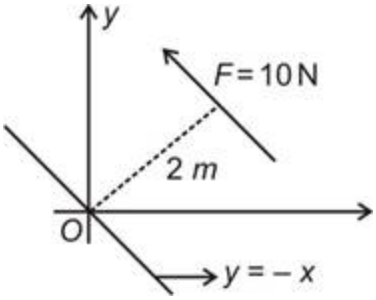
(31) **Answer :** (2)
Solution:
$\tau_{\text{net}} = 2FR + 6F \times 0 + 5F\cos 53° \times R$

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= 2*FR* + 3*FR*
= 5*FR*

(32) Answer : (4)
Solution:
There is no external force on the system. Therefore momentum of centre of mass of system remain same. It was initially rest so it will remain at rest.

(33) Answer : (1)
Solution:



Since force and axis of rotation are parallel, therefore torque of the force about axis is zero.

(34) Answer : (2)
Solution:
$k = \sqrt{\frac{I}{M}}$
$K = \sqrt{\frac{2MR^2}{5M}} \Rightarrow \frac{K^2}{R^2} = \frac{2}{5}$

(35) Answer : (1)
Solution:
$\frac{\theta}{\frac{\theta}{2^{nd}}} = \frac{3}{9} = \frac{5\theta}{\theta'} \Rightarrow \theta' = 15\theta$
$\frac{\theta}{5^{th}}$

(36) Answer : (2)
Solution:
$$\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ 1 & -2 & 1 \end{vmatrix}$$
$\bar{\tau} = (2-2)\hat{i} - (1+1)\hat{j} + (-2-2)\hat{k}$
$\vec{\tau} = 0\hat{i} - 2\hat{j} - 4\hat{k}$

(37) Answer : (4)
Solution:
$I = \frac{1}{2}MR^2$
But $M = Vd = Atd = \pi R^2 td$
$\Rightarrow R^2 = \frac{M}{\pi td}$
Now, $I = \frac{1}{2}M\frac{M}{\pi td} = \frac{M^2}{2\pi td}$
$I \propto \frac{1}{d}$
$\therefore \frac{I_A}{I_B} = \frac{d_2}{d_1}$

(38) Answer : (1)
Solution:
Net external torque $\sum \vec{\tau}_{\text{ext.}} = 0$.
$\rightarrow \vec{L} = C$
$\vec{L}_f = \vec{L}_i$
$\vec{L}_{\text{man}} + \vec{L}_{\text{platform}} = \vec{L}_i = 0$

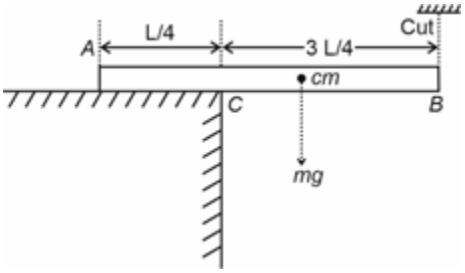
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$MvR - I\omega = 0$

$\omega = \frac{MvR}{I}$

(39) Answer : (1)

Solution:



After string B is cut point C will be instantaneous point of rotation.

Using

$T_C = I_C\alpha$

$\Rightarrow mg \times \frac{L}{4} = \left(\frac{mL^2}{12} + \frac{mL^2}{16}\right)\alpha$

$\Rightarrow \frac{mgL}{4} = \frac{7mL^2}{48}\alpha$

$\Rightarrow \alpha = \frac{12g}{7L}$

(40) Answer : (1)

Solution:

As $I_{cm} + ml_1^2 = I_{cm} + ml_2^2$

$l_1 = l_2$

(41) Answer : (2)

Solution:

10 g [45 – 12] = m g(50 – 45)

10(33) = m × 5

m = $\frac{33\times 10}{5} = 66$ grams

(42) Answer : (2)

Solution:

I = Moment of inertia before removal

I_1 = Moment of inertia of removed part about an axis perpendicular to plane and passing through O.

I' = Moment of inertia of remaining disc about an axis perpendicular to plane and passing through O.

$I' = I - I_1$

$= \frac{MR^2}{2} - \left[\frac{M}{9}\left(\frac{R/3}{2}\right)^2 + \frac{M}{9}\left(\frac{2R}{3}\right)^2\right]$

$= \frac{4MR^2}{9}$

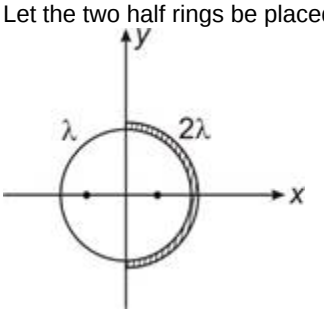
(43) Answer : (3)

Hint:

Distance of centre of mass of semicircular ring from its centre is $\frac{2R}{\pi}$.

Solution:

Let the two half rings be placed in left and right of y-axis with centre as shown in figure.



Then the coordinate of centre of mass of left and right half rings are

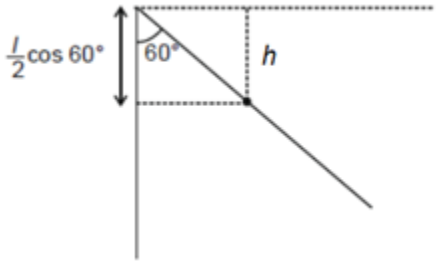
$\left(-\frac{2R}{\pi},\ 0\right)$ *and* $\left(\frac{2R}{\pi},\ 0\right)$

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$\therefore$ x-coordinates of centre of mass of complete ring is

$\frac{m\left(-\frac{2R}{\pi}\right)+2m\left(\frac{2R}{\pi}\right)}{3m}=\frac{2R}{3\pi}$

(44) **Answer :** (1)
Solution:
Gain in rotational K.E = Fall in gravitational P.E



$\frac{1}{2}\left(\frac{ml^2}{3}\right)\omega^2=mg\frac{l}{2}\cos 60^\circ$

$\Rightarrow \quad \omega=\sqrt{\frac{3g}{2l}}$

(45) **Answer :** (1)
Solution:
In elastic collision linear momentum of the system and kinetic energy of the system will conserved just before and just after the collision. But in inelastic or perfectly inelastic collision linear momentum of the system will conserved but kinetic energy will not conserved.

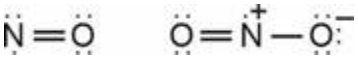
CHEMISTRY

(46) **Answer :** (4)
Hint:
Octet rule is applicable to mainly the elements of second period of periodic table.
Solution:
Octet rule does not account for the shape of molecules and it does not explain the relative stability of molecules.

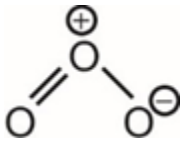
(47) **Answer :** (3)
Hint:
Molecules which have an odd number of electrons are called odd-electron molecules.

...[1 Mark]

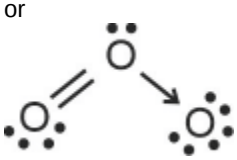
Solution:
Molecules like NO and NO_2 are odd-electron molecules.



(48) **Answer :** (3)
Solution:

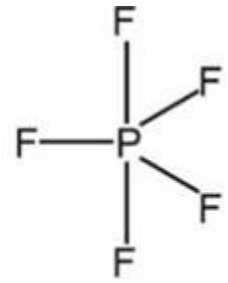


or

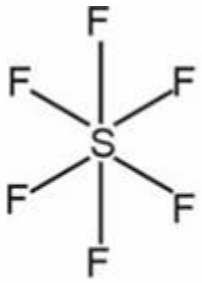


$\text{F.C}=6-\frac{1}{2}\times 6-2=6-5=+1$

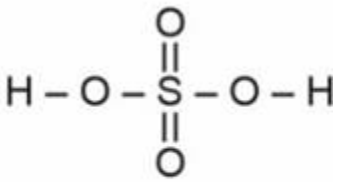
(49) **Answer :** (2)
Solution:



10 electrons around P atom



12 electrons around S atom



12 electrons around S atom

(50) **Answer :** (1)
Solution:
NaF > NaCl > NaBr > NaI: (Melting point order)

(51) **Answer :** (1)
Solution:
LiCl has highest lattice enthalpy.

(52) **Answer :** (4)
Solution:
N_2 is a pure covalent molecule.

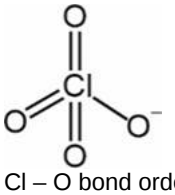
(53) **Answer :** (2)
Hint:
Bond length $\propto \frac{1}{\text{Bond order}}$
Solution:

$$\text{Bond order}: \underset{\substack{3\\(\text{Maximum})}}{CO} > \underset{2}{CO_2 = C_3O_2} > \underset{\substack{1.33\\(\text{Minimum})}}{CO_3^{2-}}$$

Therefore, maximum bond length of C – O is in CO_3^{2-}.

(54) **Answer :** (4)
Solution:
% ionic character = $\frac{2.5\times3.33\times10^{-30}\times100}{1.6\times10^{-19}\times1.5\times10^{-10}}$ = 34.7%

(55) **Answer :** (4)
Solution:



Cl – O bond order = 7/4 = 1.75

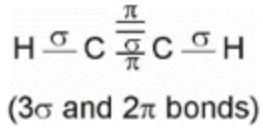
(56) **Answer :** (1)
Solution:

Bond type	Covalent bond length (pm)
C – H	107
C – N	143
C – C	154

(57) **Answer :** (1)
Solution:
Na^+ has larger size and less charge as compared to Al^{3+} and hence has more ionic nature
Cs^+ is larger in size hence it forms more ionic compound with I^- ion as compared to Na^+
Al^{3+} is smaller with more charge and hence it forms more covalent compound than Mg^{2+}

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(58) **Answer :** (3)
Solution:



(3σ and 2π bonds)

(59) **Answer :** (3)
Solution:
In case of PCl_5 axial bonds are longer than equatorial bonds.

(60) **Answer :** (4)
Solution:
In SF_4, 'S' is sp^3d hybridised and the shape of the molecule is see saw



See-saw

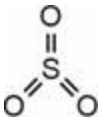
(61) **Answer :** (1)
Solution:
Correct order of ionic character
NaF > NaCl > NaBr > NaI.

(62) **Answer :** (4)
Solution:

Species	Bond Order
O_2^-	1.5
O_2	2
O_2^+	2.5

Lesser is the bond order, longer is the bond length.

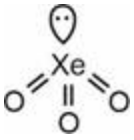
(63) **Answer :** (3)
Hint:
Shape of a molecule is determined by counting the number of bond pairs and lone pairs around central atom.
Solution:
SO_3;



sp^2 hybridised, Trigonal planar
BrF_3 ;



sp^3d hybridised, Bent T-shape
XeO_3;



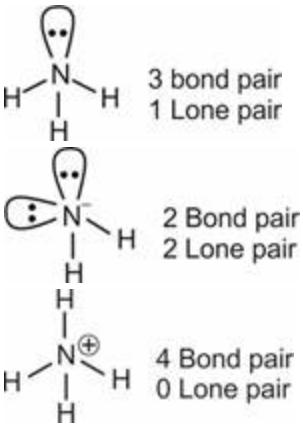
sp^3 hybridised, Trigonal pyramidal
SF_6 ;

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sp^3d^2 hybridised, Octahedral

(64) **Answer :** (2)
Hint:
Order of repulsive Interaction
lp – lp > lp – bp > bp – bp
Solution:
NH_3

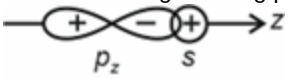


(65) **Answer :** (3)
Solution:

BF_3	120°
XeF_4	180° and 90°
PCl_5	120°, 90°, 180°
CO_2	180°

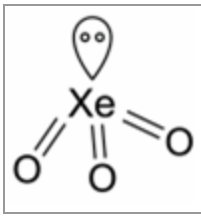
(66) **Answer :** (2)
Solution:
O = C = O carbon and oxygen uses p orbital to create π bond.

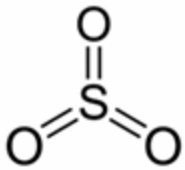
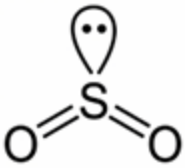
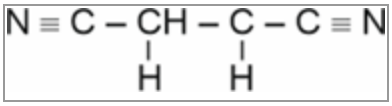
(67) **Answer :** (4)
Solution:
Orbitals forming showing positive overlap (in phase) should have same sign and orientation in space.



Out of phase overlap

(68) **Answer :** (3)
Solution:

	σ bonds	**π bonds**
	3	3
	3	3

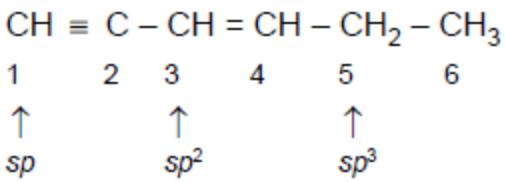
		
	2	2
	6	4

(69) **Answer :** (1)
Solution:
Mutual repulsion of the electrons in antibonding molecular orbital is more than the attraction between the electrons and the nuclei, this leads to net increase in energy. So electrons in antibonding molecular orbital destabilize the molecule.

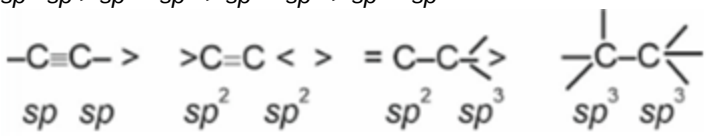
(70) **Answer :** (1)
Solution:
$\underset{sp^2}{BF_3} \rightarrow \underset{sp^3}{BF_3} \; : \; NH_3$

(71) **Answer :** (3)
Solution:
The hybrid orbitals are more effective in forming stable bonds then the pure atomic orbitals.

(72) **Answer :** (3)
Solution:



(73) **Answer :** (4)
Solution:
Bond energy order:
sp - *sp* > sp^2 - sp^2 > sp^2 - sp^3 > sp^3 - sp^3



(74) **Answer :** (3)
Solution:
Approximate percentage of p character in sp^2 hybridisation is 66% – p character, 33% – s – character

(75) **Answer :** (4)
Solution:



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(76) **Answer :** (2)
Solution:

Molecular Orbital Configuration of O_2 ($16e^-$) is

$\sigma 1s^2, \sigma^*1s^2, \sigma 2s^2, \sigma^*2s^2, \sigma 2p_z{}^2, \begin{cases}\pi 2p_x^2 \\ \pi 2p_y^2\end{cases}, \begin{cases}\pi^* 2p_x^1 \\ \pi^* 2p_y^1\end{cases}$

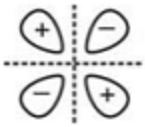
(77) **Answer :** (1)
Solution:
If bond order is zero, the species does not exist.

(78) **Answer :** (4)
Solution:

Electronic configuration of He_2^+ : $\sigma 1s^2 \sigma^* 1s^1$

(79) **Answer :** (1)
Solution:
H_2^- is less stable than H_2^+ because in $H_2^\ominus$ electron enters in antibonding (σ^*1s) molecular orbital.
$\because$ Order of BDE is $H_2^- < H_2^+ < H_2$

(80) **Answer :** (2)
Solution:



represents π-antibonding molecular orbital.

(81) **Answer :** (2)
Solution:
During $N_2 \rightarrow N_2^-$ ion, electron will add up to $\pi^* 2p_x$ or $\pi^* 2p_y$ orbital

(82) **Answer :** (2)
Solution:
In C_2 only π bonds are present

(83) **Answer :** (4)
Solution:
O_2^{2-} does not contain unpaired electron.

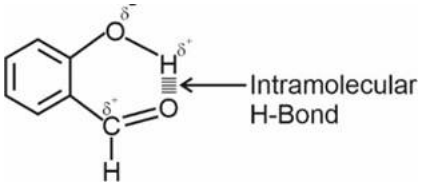
(84) **Answer :** (1)
Solution:

Molecule	Bond order	Bond length (pm)
Li_2	1	267.2
F_2	1	141.8

(85) **Answer :** (4)
Solution:
If x axis is intermolecular axis, then 2*py* and 2*py* will not form σ bond.

(86) **Answer :** (2)
Solution:
H atom bonded with F, O and N generally gets involved in H-bonding.

(87) **Answer :** (2)
Solution:

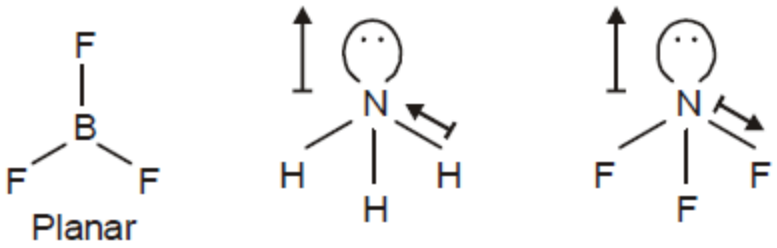


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(88) **Answer :** (3)
Solution:
In NH_4^+, nitrogen has 4 bond pairs only presents in *s* and *p* orbitals therefore hybridisation of N is sp^3.

(89) **Answer :** (4)
Solution:
CO_3^{2-} → 32 electrons, sp^2
NO_3^- → 32 electrons, sp^2
ClO_3^- → 42 electrons, sp^3
SO_3^{2-} → 42 electrons, sp^3

(90) **Answer :** (2)
Solution:



BOTANY

(91) **Answer :** (4)
Solution:
Mycoplasma is the smallest wall-less moneran, which is mostly parasitic.

(92) **Answer :** (4)
Hint:
This organism is *Nostoc.*
Solution:
Nostoc is a cyanobacterium that can fix both atmospheric nitrogen and carbon dioxide.

(93) **Answer :** (2)
Solution:
Viruses did not find a place in classification, since they are not considered truly 'living', if we understand living as those organisms that have a cell structure.

(94) **Answer :** (3)
Solution:
Viroids caused potato spindle tuber disease.

(95) **Answer :** (4)
Solution:
Lichens do not grow in polluted areas.

(96) **Answer :** (1)
Solution:
Karyogamy involve fusion of two nuclei.
Plasmogamy is fusion of protoplasm.
Meiosis leads to the production of haploid spores.

(97) **Answer :** (3)
Solution:
Kingdom Monera includes all prokaryotes and they show many different types of mode of nutrition.

(98) **Answer :** (2)
Solution:
The vast majority of bacteria are heterotrophs.

(99) **Answer :** (4)

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Solution:
Diatoms are chief producers in the oceans. Their cell wall is impregnated with silica to form transparent siliceous shell, known as frustule.

(100) Answer : (3)
Solution:
Agaricus (Mushrooms) forms basidiocarps which are edible.

(101) Answer : (3)
Solution:
Conidia are asexual spores.

(102) Answer : (3)
Solution:
Dinoflagellates contain two flagella, one is longitudinal and the other is transverse.

(103) Answer : (4)
Solution:
Slime moulds produce plasmodium under favourable condition.

(104) Answer : (3)
Solution:
Spherical shaped bacteria are called "**Cocci**".

(105) Answer : (4)
Solution:
Prions are abnormally folded proteins. They entirely lack nucleic acid.

(106) Answer : (4)
Hint:
Spirogyra is not a unicellular organism.
Solution:
Kingdom Protista has brought together *Chlamydomonas, Chlorella* with *Paramoecium* and *Amoeba*.

(107) Answer : (2)
Solution:
Viroids are free RNA; it lacks the protein coat that is found in viruses. Mycorrhizae are a symbiotic association between higher plant roots and fungi. Viruses contain genetic material that could be either RNA or DNA. Prions are abnormally folded proteins and are similar in size to viruses.

(108) Answer : (3)
Solution:
Euglena is chlorophyllous, flagellate protist.

(109) Answer : (3)
Solution:
Tetanus is caused by *Clostridium tetani* (bacterium).

(110) Answer : (1)
Solution:
TMV genetic material is ssRNA.

(111) Answer : (3)
Hint:
Saccharomyces are unicellular fungi.
Solution:
With the exception of yeasts which are unicellular, fungi are filamentous with aseptate or septate hyphae.

(112) Answer : (3)
Solution:
Gonyaulax has a cell wall that has stiff cellulosic plates on the outer surface.

(113) Answer : (4)
Solution:
Paramoecium and *Plasmodium* both are unicellular.
Both belong to the group Protozoa.

(114) Answer : (4)
Solution:
Albugo acts as a parasite and causes white rust on mustard.

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(115) Answer : (3)
Hint:
Methanogens are found in marshy areas.
Solution:
Methanogens are obligate anaerobes and chemoautotrophs.

(116) Answer : (2)
Hint:
Deuteromycetes lack perfect stage in their life cycle.
Solution:
Deuteromycetes do not reproduce sexually.

(117) Answer : (3)
Solution:
Trypanosoma causes sleeping sickness. It is a flagellated protozoan.

(118) Answer : (1)
Solution:
In five kingdom classification, unicellular eukaryotic organisms are placed in kingdom Protista.

(119) Answer : (2)
Solution:
In *Nostoc* nitrogen fixation occurs in thick walled heterocysts.

(120) Answer : (2)
Solution:
Aristotle used simple morphological characters to classify plants into herbs, shrubs and trees.
Herbs are non-woody plants.
Shrubs are taller than herbs and their stems are not harder but thicker. Trees are woody plants taller than herb and shrub.

(121) Answer : (3)
Solution:
In two kingdom classification, system *Chlamydomonas* and *Spirogyra* were included in kingdom plantae.

(122) Answer : (2)
Solution:
Some hyphae are continuous tubes filled with multinucleated cytoplasm and are known as coenocytic hyphae. Coenocytic mycelium is present in the members of Phycomycetes *e.g*., *Rhizopus*, *Mucor* and *Albugo.*

(123) Answer : (2)
Hint:
Paramoecium is a ciliated protozoan.
Solution:
In *Paramoecium* the coordinated movement of rows of cilia causes the water laden with food to be steered into the gullet.
With the exception of yeasts which are unicellular, fungi are filamentous.

(124) Answer : (4)
Solution:
Sleeping sickness causing organism is *Trypanosoma.*

(125) Answer : (4)
Solution:
M.W. Beijerinek called the infectious living fluid as *Contagium vivum fluidum*.

(126) Answer : (2)
Solution:
Neurospora is extensively used in biochemical and genetic work.

(127) Answer : (3)
Solution:
Red dinoflagellates*, e.g., Gonyaulax,* undergo rapid multiplication that makes the sea appear red (red tides).

(128) Answer : (4)
Solution:
Rusts and smuts causing organisms belong to Basidiomycetes.

(129) Answer : (1)
Solution:
Glycogen is reserve food in Fungi.

(130) Answer : (3)

Hint:
Kuru is a neurological diseases that is transmitted by an agent consisting of abnormally folded protein.
Solution:
Kuru disease is caused by prions.

(131) Answer : (3)
Solution:
Loose tissue body organisation is found in fungi.

(132) Answer : (3)
Solution:
Slime moulds show both asexual as well as sexual reproduction.

(133) Answer : (1)
Solution:
In five kingdom classification, all unicellular eukaryotes are placed in kingdom Protista.

(134) Answer : (2)
Solution:
Cuscuta (Amar bel) is a parasitic plant.

(135) Answer : (3)
Solution:
Viruses are obligate parasites.

ZOOLOGY

(136) Answer : (2)
Solution:

•	Cardiac arrest	–	Heart stops beating
•	Heart failure	–	Heart fails to pump blood effectively. It is sometimes called congestive heart failure because congestion of the lungs is one of the main symptoms of this disease.

(137) Answer : (1)
Solution:
In emphysema, alveolar wall becomes damaged decreasing vital capacity of lungs. Silicosis and asbestosis are occupational lung diseases causing serious lung damage due to inflammation and fibrosis.

(138) Answer : (3)
Solution:
Blood normally contains 1,50,000 - 3,50,000 platelets mm^{-3} and 6,000 – 8,000 WBCs mm^{-3} of blood.

(139) Answer : (4)
Solution:
In amphibians and reptiles, heart pumps out mixed blood, thus, they exhibit incomplete double circulation.

(140) Answer : (2)
Solution:
First heart sound – Closure of the bicuspid and tricuspid valves.
Second heart sound – Closure of the semilunar valves.

(141) Answer : (3)
Solution:
The SAN can generate maximum number of action potentials and is responsible for initiating and maintaining rhythmic contractile activity of heart. Therefore, it is called the pacemaker. It is situated in right upper corner of right atrium. AV node is situated in lower left corner of right atrium close to the atrio-ventricular septum.

(142) Answer : (1)
Hint:
Thrombocytes
Solution:
Platelets, also called thrombocytes, are cell fragments produced from megakaryocytes. A reduction in their number can lead to clotting disorders.

Erythrocytes have a red coloured, iron containing complex protein called haemoglobin. Haemoglobin plays a significant role in transport of respiratory gases.
Monocytes are phagocytic WBCs.

(143) Answer : (4)
Solution:
All the four chambers of heart are in a relaxed state*, i.e.,* they are in joint diastole. As the tricuspid and bicuspid valves are open, blood from the pulmonary veins and vena cava flows into the left and the right ventricles, respectively, through the respective left and right atria. The semilunar valves are closed at this stage. The SAN now generates an action potential which stimulates both the atria to undergo a simultaneous contraction, known as the atrial systole. This increases the flow of blood into the ventricles by about 30 per cent.

(144) Answer : (3)
Solution:
Oxygen hemoglobin dissociation curve is sigmoid or S-shaped.

(145) Answer : (3)
Solution:
Under normal conditions, 100 mL of oxygenated blood delivers → 5 mL of O_2 to tissues.

$\therefore$ 5 L will deliver → $\frac{5\times5000}{100} = 250\,mL$

(146) Answer : (2)
Solution:

$$\left.\begin{matrix} pO_2 = 95\ mmHg \\ pCO_2 = 40\ mmHg \end{matrix}\right\} \text{Oxygenated blood}$$

(147) Answer : (2)
Solution:
Fibrins are formed by the conversion of inactive fibrinogens in the plasma by the enzyme thrombin. Thrombins, in turn are formed from another inactive substance present in the plasma called prothrombin. An enzyme complex, thrombokinase, is required for the above reaction. This complex is formed by a series of linked enzymatic reactions (cascade process) involving a number of factors present in the plasma in an inactive state. An injury or a trauma stimulates the platelets in the blood to release certain factors which activate the mechanism of coagulation.

(148) Answer : (4)
Solution:
The contraction of the ventricle starts shortly after Q and marks the beginning of the ventricular systole.

(149) Answer : (3)
Solution:
Plasma without clotting factors is called serum.

(150) Answer : (4)
Solution:

pO_2 (mmHg)	pCO_2 (mmHg)
Systemic artery $= 95$	Tissues $= 45$
Pulmonary artery $= 40$	Pulmonary vein $= 40$
	Atmospheric air $= 0.3$

(151) Answer : (1)
Solution:
If blood pressure of an individual is 140/90 mmHg or higher even after repeated checks, it shows hypertension. High blood pressure leads to heart diseases and also affects vital organs like brain and kidney.

(152) Answer : (2)
Solution:
Cardiac arrest is a condition in which the heart stops beating.

(153) Answer : (2)
Solution:
The wall of alveoli is lined by single layered squamous epithelium and alveolar capillaries is lined by single layered endothelium and in between these layers, acellular basement substance is present. So, oxygen crosses two cellular layers and one acellular layer.

(154) Answer : (1)
Solution:

Bicuspid valve is also known as mitral valve which is present between left atrium and left ventricle.
Tricuspid valve is present between right atrium and right ventricle.

(155) Answer : (3)
Solution:
The O^{-ve} blood group individuals are universal donors while AB^{+ve} blood group individuals are universal recipients.
An Rh^{-ve} person, if exposed to Rh^{+ve} blood, will form specific antibodies against the Rh antigen thereby causing clumping of RBCs.

(156) Answer : (1)
Solution:
Since, the human heart is autoregulated by the patches of nodal tissues (SA node, AV node) which are made of cardiac muscles, it is said to be myogenic in nature.

(157) Answer : (3)
Solution:
QRS complex indicate ventricular depolarisation hence number of QRS complexes represents number of heartbeats.

(158) Answer : (2)
Solution:
Transport of CO_2
(a) 7% in dissolved form in blood plasma
(b) 23% as carbamino-haemoglobin
(c) 70% as bicarbonate ion in blood plasma

(159) Answer : (3)
Solution:
Increase in CO_2 and H^+ concentration can activate the chemosensitive area which in turn can signal the respiratory rhythm centre to make necessary arrangements in the respiratory process by which these substances can be eliminated.

(160) Answer : (1)
Solution:
In industries that involve grinding or stone-breaking, so much dust is produced that the defense mechanism of the body cannot fully cope with the situation. Long exposure can give rise to inflammation leading to fibrosis (proliferation of fibrous tissues) and thus causing serious lung damage.

(161) Answer : (1)
Solution:
ABO grouping is based on the presence or absence of two surface antigens on the RBCs namely A and B.

(162) Answer : (4)
Solution:
O_2 and CO_2 are exchanged at alveoli by simple diffusion mainly based on pressure/concentration gradient.

(163) Answer : (2)
Solution:
Hint : It is equal to the average number of heart beats per minute.
Sol. : The SAN can generate maximum number of action potentials of 70-75/minute in comparison to any other nodal tissues.

(164) Answer : (3)
Solution:
During ventricular systole, ventricular pressure increases and semilunar valves are forced open.

(165) Answer : (1)
Solution:
Carboxypeptidase is a pancreatic enzyme. Higher amount of carbonic anhydrase is present in RBC.

(166) Answer : (2)
Solution:
Atherosclerosis is caused due to deposition of cholesterol fat, calcium and fibrous tissue.

(167) Answer : (4)
Solution:
P wave = Contraction of both atria
QRS Complex = Depolarisation of ventricles

(168) Answer : (3)
Solution:
Artery usually carries blood away from the heart, whereas vein usually carries blood towards the heart.

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(169) Answer : (3)
Solution:
SAN (Sino-atrial node) is called the pacemaker of the human heart that initiates and generates maximum action potentials.

(170) Answer : (3)
Solution:
Every 100 mL of oxygenated blood can deliver around 5 mL of O_2 to the tissues and 4 mL of CO_2 to the alveoli under normal physiological conditions in humans.

(171) Answer : (3)
Solution:
The solubility of CO_2 is 20-25 times higher than that of O_2, so the amount of CO_2 that can diffuse through the diffusion membrane per unit difference in partial pressure is much higher as compared to that of O_2.

(172) Answer : (3)
Solution:
Platelets are actually cell fragments.

(173) Answer : (4)
Solution:
Hint : Normal haemoglobin level.
Sol. : The average life span of an RBC is 120 days, after which they are destroyed in the spleen (graveyard of RBCs). RBCs contain haemoglobin, which helps in carrying oxygen. A healthy individual has 12-16 g of haemoglobin in every 100 mL of blood.

(174) Answer : (2)
Solution:
Tissues are known as 'CO_2 producing factory' because they produce CO_2 after performing cellular respiration. The pCO_2 in tissues is found to be 45 mmHg.

(175) Answer : (2)
Solution:
97% of O_2 is transported by RBCs in the blood as oxyhaemoglobin.

(176) Answer : (3)
Solution:
Crocodiles are reptiles with four-chambered heart. Amphibians have three-chambered heart.

(177) Answer : (1)
Solution:
Erythrocytes or red blood cells are the most abundant of all the cells in blood *i.e.*, 5-5.5 million mm^{-3} of blood.
* Leucocytes – 6000-8000 mm^{-3} of blood
* Thrombocytes or platelets – 1,50,000 – 3,50,000 mm^{-3} of blood

(178) Answer : (2)
Solution:
Medulla oblongata can moderate the cardiac functions through autonomic nervous system. Stimulation of sympathetic nerve can increase the rate of heart beat and neural signal through parasympathetic system decreases the rate of heart beat.

(179) Answer : (4)
Solution:
Shifting of O_2 – dissociation curve towards left indicates the association of O_2 and Hb. Following are the conditions responsible for this shifting :
(i) High pO_2
(ii) Low pCO_2
(iii) Less H^+ ion concentration
(iv) Low temperature

(180) Answer : (1)
Solution:
Silicosis is an example of occupational respiratory disorder. Pneumonia is a lung disease characterized by inflammation of air spaces in lungs and most commonly occur due to a bacterial infection.

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